Quantum Algorithms versus Classical Algorithms: A Literature-Based Research Paper

Abstract

Quantum algorithms promise speedups over classical ones, but the promise is narrow and often misunderstood. This paper reviews fourteen studies on quantum and classical algorithms, grouped into five themes: oracle separations and foundations, factoring and search, linear algebra and dequantization, near-term hybrid methods, and advantage claims with resource estimates. Each theme is analyzed for the problem addressed, the algorithm, the reported speedup, and its limitations. A comparative table is included. The review shows that speedups range from exponential (factoring, some oracle problems) to quadratic (search) to none (most everyday tasks), that several claimed speedups disappeared once better classical algorithms were found, and that hardware noise is the main barrier to practical use. It concludes with open challenges: fault tolerance, fair benchmarking, input and output costs, and the pace of classical improvement.

1. Introduction

A classical computer stores information in bits, each 0 or 1. A quantum computer stores it in qubits, which can be in a superposition of 0 and 1. With n qubits, the state is described by 2^n amplitudes. Algorithms can use interference to make correct answers more likely when the state is measured.

That description invites a tempting claim: a quantum computer tries every answer at once. The claim is wrong. Measurement returns one outcome, so a useful algorithm has to arrange amplitudes so that the right answer dominates. Very few problems are known where this can be done faster than classically.

This paper asks two questions. Which problems have quantum algorithms that provably or plausibly beat classical ones, and where does the advantage vanish or remain unproven? To answer them, it reviews fourteen papers chosen to cover different problem types rather than to be exhaustive.

Section 2 introduces the concepts used later. Section 3 reviews the literature by theme. Section 4 compares the studies. The remaining sections cover open challenges and conclusions.

2. Background

Classical algorithms run on deterministic or randomized machines. Their cost is measured as time or queries as a function of input size n. Complexity classes include P (efficient deterministic), BPP (efficient randomized) and NP (efficiently checkable solutions).

Quantum algorithms run on circuits of unitary gates applied to qubits. Three ideas matter. Superposition lets a register hold many basis states at once. Entanglement links qubits so they cannot be described separately. Interference makes amplitudes add or cancel, which is where speedups come from. The class BQP contains problems solvable efficiently by a quantum computer with bounded error. It is known that P is contained in BQP, but how BQP relates to NP is open.

Two kinds of speedup appear in the literature. A query separation counts calls to a black-box oracle, which can be proven but may not reflect real inputs. A time-complexity speedup counts total running time and depends on the best known classical competitor, which can improve over time.

Practical limits apply to every quantum algorithm. Qubits suffer noise and decoherence. Error correction needs many physical qubits per logical qubit. Loading classical data into a quantum state, and reading out answers, can cost as much as the speedup saves. Preskill called the current era NISQ (noisy intermediate-scale quantum).

3. Literature Review

The fourteen papers fall into five overlapping groups. Each subsection gives the problem, the method, the reported findings and the limitations.

3.1 Oracle Separations and Foundations

Deutsch and Jozsa (1992) gave the first problem where a quantum algorithm needs fewer oracle queries than any deterministic classical one. Their task is to decide whether a hidden function on n bits is constant or balanced. In its modern form a quantum computer needs one query, while a deterministic classical algorithm may need 2^(n-1) + 1. The advantage is real but weak, because a randomized classical algorithm solves the problem with high probability in a constant number of queries.

Bernstein and Vazirani (1997) formalized quantum complexity theory, defined BQP, and gave an oracle problem separating quantum from classical probabilistic computation. Simon (1997) went further with a problem that quantum computers solve in polynomially many queries while any classical randomized algorithm needs exponentially many. Simon's period-finding structure directly inspired Shor's algorithm.

The limitation of this group is that oracles are abstract. A separation relative to an oracle shows that quantum computing can be stronger, but it does not by itself give a useful algorithm on real inputs.

3.2 Factoring and Search

Shor (1994) gave a quantum algorithm that factors an integer N in time polynomial in log N, using period finding and the quantum Fourier transform. The best known classical method, the general number field sieve, runs in sub-exponential time. Shor's result is the strongest known evidence of a useful exponential speedup and is the reason post-quantum cryptography exists. It is not proven that no fast classical factoring algorithm exists, only that none is known.

Grover (1996) gave a quantum algorithm that searches an unstructured collection of N items in about √N steps using amplitude amplification, where a classical algorithm needs on the order of N. Bennett, Bernstein, Brassard and Vazirani (1997) proved that this is optimal, so no quantum algorithm can do better than a quadratic speedup for unstructured search. Grover's speedup is therefore broad but modest. It also assumes the search can be expressed as an efficient oracle.

3.3 Linear Algebra and Dequantization

Harrow, Hassidim and Lloyd (2009), known as HHL, gave a quantum algorithm for solving linear systems that can run exponentially faster than classical methods when the matrix is sparse and well-conditioned and only a property of the solution is needed. It became the basis for many quantum machine learning proposals.

Aaronson (2015) examined these claims and listed the fine print: the input must be loaded efficiently, the output is a quantum state rather than a full vector, and conditioning must be good. When these conditions fail, the advantage disappears.

Tang (2019) then showed that for the recommendation-systems problem, which had been presented as an exponential quantum speedup, a classical algorithm with comparable sampling access to the data runs in time polylogarithmic in the dimensions. This removed the claimed exponential advantage for that problem. The lesson is that a quantum speedup is only as strong as the classical algorithm it is compared against.

3.4 Near-Term Hybrid Algorithms

Peruzzo and colleagues (2014) introduced the variational quantum eigensolver (VQE) on a photonic processor. A short quantum circuit prepares a trial state, and a classical optimizer adjusts its parameters to minimize an energy estimate. The design suits small, noisy devices. Farhi, Goldstone and Gutmann (2014) proposed the quantum approximate optimization algorithm (QAOA) for combinatorial problems, using the same quantum-classical loop.

Preskill (2018) framed the surrounding era as NISQ and argued that such devices may be scientifically useful but are not yet able to run algorithms like Shor's. For hybrid methods there is no proof of an advantage over the best classical heuristics. Performance depends on circuit depth, noise and optimizer behavior.

3.5 Advantage Claims and Resource Estimates

Arute and colleagues (2019) reported a 53-qubit processor completing a random circuit sampling task in about 200 seconds, which they estimated would take a classical supercomputer thousands of years. Later work using improved classical simulation techniques substantially reduced the classical estimate. The task also has no practical use, so the result is best read as a hardware milestone rather than an algorithmic win.

Gidney and Ekerå (2021) estimated the resources needed to break RSA-2048 with Shor's algorithm: about 20 million noisy physical qubits and roughly eight hours, under stated assumptions about error rates and error correction. That is far beyond current machines, which gives a concrete measure of the distance between theory and practice.

4. Comparative Discussion

The fourteen papers differ most on three things: what kind of speedup is claimed, how strong the evidence is, and what practical conditions the speedup requires. Table 1 summarizes each paper.

Table 1. Summary of the reviewed studies

#

Paper

Problem

Quantum result

Classical comparison

Main limitation

1

Deutsch and Jozsa (1992)

Constant vs balanced function

1 query

2^(n-1)+1 queries (deterministic)

Randomized classical needs O(1) queries

2

Bernstein and Vazirani (1997)

Quantum complexity theory

Defined BQP, oracle separation

Probabilistic classical

Oracle setting only

3

Simon (1997)

Hidden period

Polynomial queries

Exponential queries

Oracle setting, artificial problem

4

Shor (1994)

Integer factoring

Polynomial in log N

Sub-exponential (GNFS)

Needs fault-tolerant hardware

5

Grover (1996)

Unstructured search

O(√N)

O(N)

Only quadratic, needs efficient oracle

6

Bennett et al. (1997)

Limits of quantum search

Proves √N is optimal

O(N)

Negative result, no new speedup

7

Harrow et al. (2009)

Linear systems

Polylog in dimension

Polynomial

Strong input and output conditions

8

Peruzzo et al. (2014)

Ground-state energy (VQE)

Hybrid circuit plus optimizer

Classical chemistry methods

No proven advantage

9

Farhi et al. (2014)

Combinatorial optimization (QAOA)

Hybrid approximation

Classical heuristics

No proven advantage

10

Preskill (2018)

NISQ-era outlook

Perspective

Not applicable

Position paper, no experiments

11

Arute et al. (2019)

Random circuit sampling

About 200 seconds

Estimated thousands of years

Classical estimate later reduced, no practical use

12

Tang (2019)

Recommendation systems

Dequantized

Classical polylog algorithm

Removes a claimed speedup

13

Gidney and Ekerå (2021)

Resources to factor RSA-2048

About 20 million qubits, 8 hours

Not applicable

Depends on assumed error rates

14

Aaronson (2015)

Fine print of quantum machine learning

Analysis

Not applicable

Commentary, no new algorithm

The first point is that speedups come in three strengths. Factoring and some oracle problems are exponential, search is quadratic, and for most tasks, including sorting and general database queries, no advantage is known. Treating all quantum algorithms as exponentially faster is the most common mistake.

The second point is that the classical baseline moves. Tang's result and the revised estimate for random circuit sampling show that a claimed advantage can shrink or vanish when classical researchers respond. The fair comparison is against the best classical algorithm, not the first one tried.

The third point is that proofs and practice are far apart. Oracle separations are proven but abstract. Shor's algorithm is practical in theory but needs millions of qubits. Hybrid methods run today but come with no proof of advantage.

5. Open Challenges

First is fault tolerance. Current devices are noisy, and error correction multiplies the number of physical qubits needed. Until logical qubits are reliable at scale, algorithms like Shor's remain theoretical.

Second is fair benchmarking. Papers report speedups against different classical baselines and on different problems, so results cannot be ranked against each other. Shared benchmarks and clear reporting of the strongest classical competitor are needed.

Third is data input and output. Many algorithms assume data is already in a quantum state, and many produce quantum states that are costly to read out. These costs can cancel the speedup, as Aaronson's analysis shows.

Fourth is the pace of classical improvement. Dequantization shows that classical algorithms can catch up, so any claimed advantage has to be treated as provisional.

Fifth is the lack of evidence for near-term advantage. Variational and optimization algorithms are promising but unproven, and finding problems where a noisy machine truly wins is still open.

Lastly, cryptography must prepare early. Because encrypted data can be stored now and decrypted later, migration to post-quantum cryptography is urgent even though large machines are years away.

6. Conclusion

Quantum algorithms are powerful for a small set of problems and unremarkable for the rest. Shor's algorithm offers an exponential speedup for factoring, Grover's offers a quadratic one for search, and many proposed speedups, especially in machine learning, depend on conditions that are hard to meet or have been matched by classical algorithms. Classical computing remains the right tool for nearly all practical work, and quantum computers are best seen as specialized accelerators for specific problems.

This review has limits. The fourteen papers cover different problem types but not the whole literature, and they exclude recent advances in error-correcting codes and hardware after 2021. Bibliographic details were written from memory of the original publications and should be checked against the source pages before any formal submission.

A logical next step is to implement Grover's and a small variational algorithm on a simulator, compare them with classical baselines on the same problem, and record how noise changes the results. The pivotal question remains: for a given problem, is the quantum speedup real, is it needed, and has the best classical algorithm been tried?

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